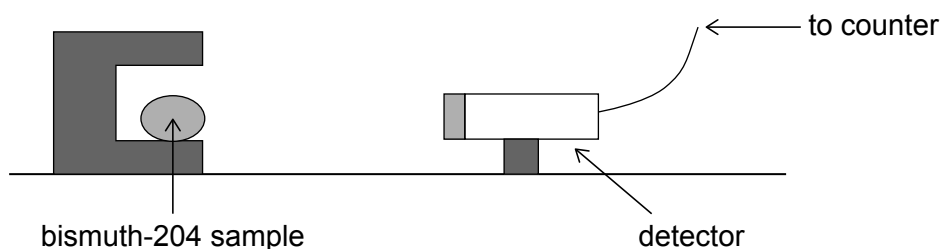


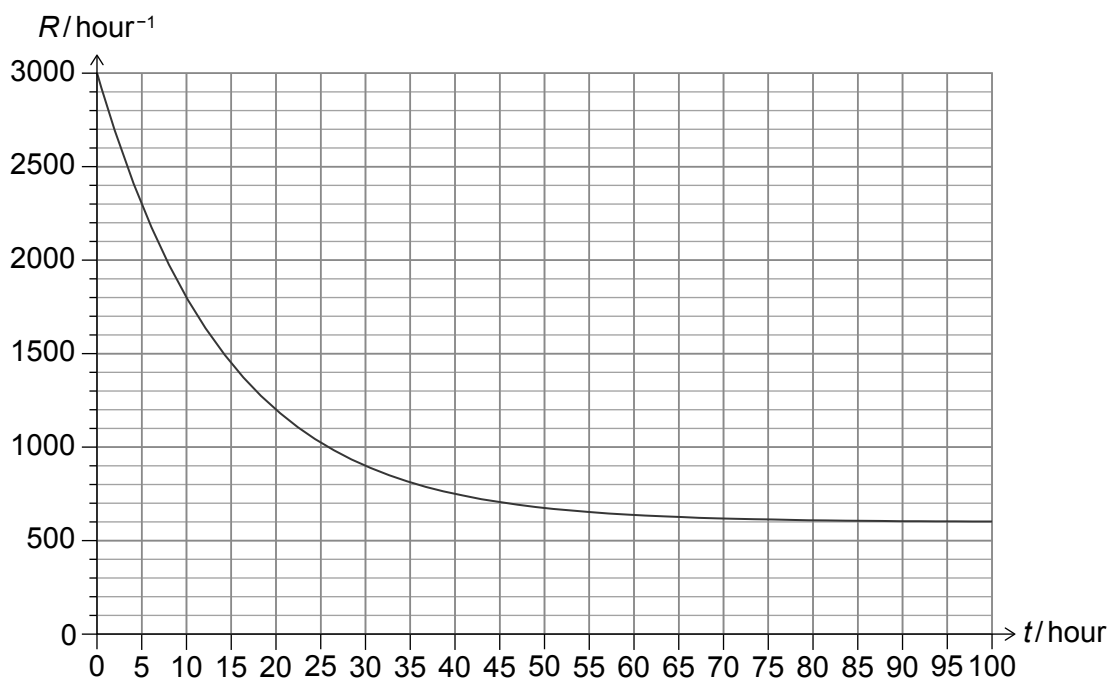
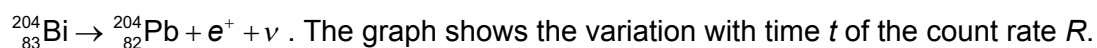
E.3 / PAPER 2 / QUESTION 3

6 November 2025 / TZ3 (file label) / HL Paper 2 / original Q4 / 8825-6721

3. (a) A pure radioactive sample of bismuth-204 ($^{204}_{83}\text{Bi}$) is placed near a detector, which is connected to a counter.



Bismuth-204 decays into lead ($^{204}_{82}\text{Pb}$) by beta-plus decay according to the equation



- (i) State the background count rate.

[1]

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(This question continues on the following page)

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(Question 3 continued)

(ii) Estimate the half-life of $^{204}_{83}\text{Bi}$.

[3]

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(iii) State and explain how the binding energy per nucleon of $^{204}_{83}\text{Bi}$ compares with that of $^{204}_{82}\text{Pb}$.

[2]

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(iv) The distance between the sample and the detector is 20 cm. State why this would be an inappropriate distance for a sample that decayed by alpha decay.

[1]

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(b) The following data are available for **nuclear** masses:

bismuth = 203.932254 u

lead = 203.928034 u

Calculate the energy released in the decay of one nucleus of bismuth-204.

[2]

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